NYU | TANDON

MODULE 3 · INTELLIGENT AUTOMATION + AI AGENTS

From RPA to Intelligent Automation

Last week your bot broke at the judgment step. This week you meet what takes it.

Friday, October 16, 2026 · 8:00–10:30 AM · 2 MTC Room 802

IE-GY 9113B — Systems Integration: From ERP to Agentic AI · Prof. Emanuele Cimica

TODAY

8:00

What this morning is for

Warm-up + recall	8:05	Rebuild the spine together — where we are on the ladder, and the step a rule can't take.
The bank that learned to read	8:25	JPMorgan COiN — a machine takes a judgment step 360,000 hours wide.
Your team's work	8:55	Name the AI-judgment step in your own process → Board Memo §5, slide S8.
Industry panel (mid-semester)	9:45	A step back with our partners across the whole course — integrate to agentic, and the board panel ahead.
Wrap + what's next	10:20	Your deliverable, and the SPAR loop you meet next week.

What you walk out with: the exact piece your board deck needs next (slide S8), plus a practitioner's read on the whole journey you're building toward.

BEFORE WE START

A bank was spending this much, every year, reading its own contracts:

360,000

lawyer- & loan-officer hours a year

12,000 contracts a year · every one read and interpreted by hand · **then: seconds, by machine**

Today is about the step a rule can't take, and who is accountable when a machine takes it.

Son, Bloomberg (2017); open access via ABA Journal (2017).

RECALL · YOUR TURN

8:05

Where are we on the ladder — and what does this week add?

Last week you ran a bot. Which rung were you on, and which one are we climbing to now?

integrate

automate

make intelligent

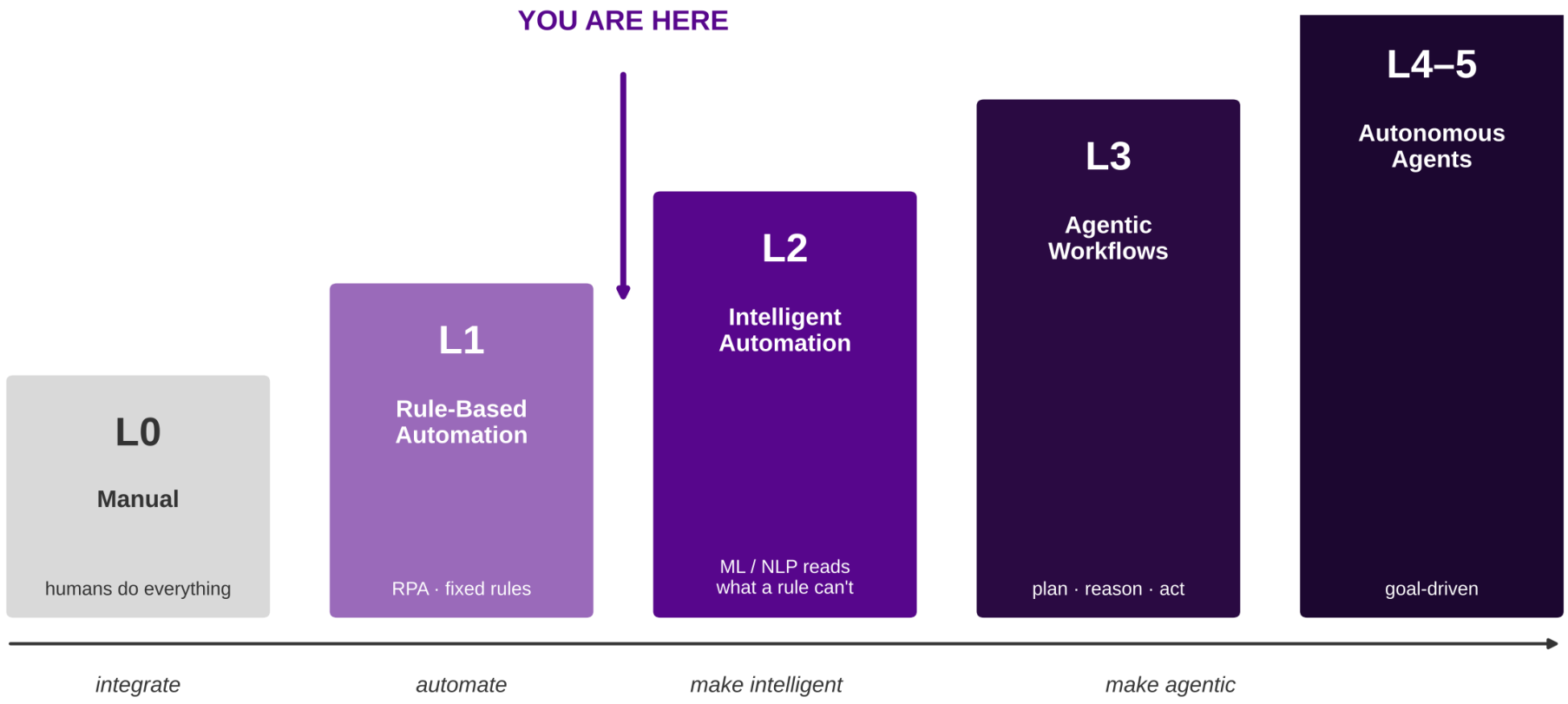
make agentic

Quick one — I ask, you answer. No huddle; hands up and we move straight on.

RECALL · REVEAL

8:08

The step up to intelligent automation



You are not leaving RPA behind. You add the rung above it, for the steps a rule can't hold. Same backbone, same bots, one new layer.

RECALL · YOUR TURN

8:10

Two kinds of decision point — which gets a bot, which gets a brain?

You have been sorting these since Lab 1, on your process map and in what you chose to automate. Recall the logic, then point to where you already met both.

The logic

What makes a decision point stable-rules versus judgment? One line for each.

In your own work

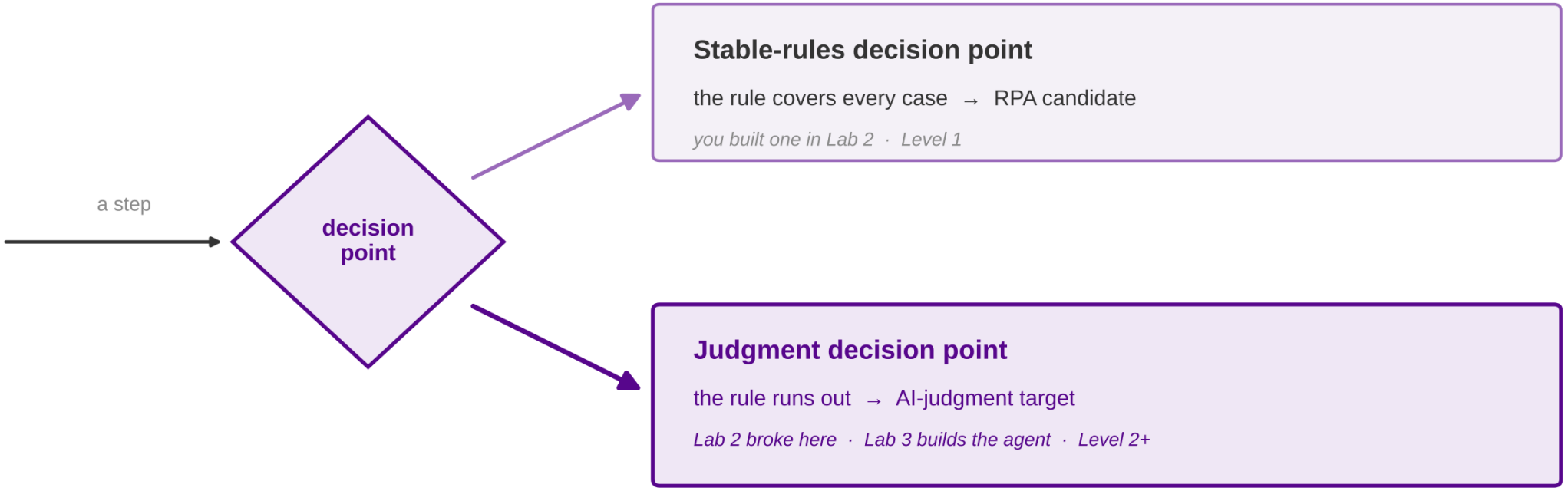
Where on your map, or in a breakout, did you tag each kind — and how did it shape what you automated?

Talk it out in your team — 60 seconds. Then we reveal and sharpen it together.

RECALL · REVEAL

8:13

Every decision point splits two ways



You have been doing this since Lab 1: tagging points on your map and choosing automations around them. The costly pattern is a judgment point treated as a stable one, where the rule fires with full confidence and is wrong.

RECALL · YOUR TURN

8:15

A rule can't read a contract. What can — and in how many ways?

Name the kind of system that reads prose, images, and clauses, and its three sub-types, one example each.

The problem

A rule needs a value in a field. A contract is prose — meaning with no field to point at.

Your recall

What reads it? And what are the three ways such a system learns?

Talk it out in your team — 60 seconds. Then we reveal and sharpen it together.

RECALL · REVEAL

8:18

A rule can't read a contract — three ways a machine can

What reads it: machine learning (ML). For text like a contract, that is natural-language processing (NLP); its most capable form today is a large language model (LLM). ML learns in three ways:

Supervised

Learns from an expert's labels — reads a contract the way a lawyer marked up thousands before it.

Unsupervised

Finds the odd one out — the transaction or record that doesn't fit the pattern. Anomaly detection.

Reinforcement

Learns by trying toward a goal — keeps what improves its position, discards what doesn't.

Intelligent automation bolts a model onto a rule-based flow: the model takes the judgment step; the rule keeps the structured remainder.

RECALL · YOUR TURN

8:20

RPA sells on cost. What does intelligent automation sell?

A bot is a cheaper clerk. What does a machine that reads all of the work add?

Rule-based bot → ?

What is the business case — in a word?

Intelligent automation → ?

What is the bigger prize, once a machine reads everything, not a sample?

Talk it out in your team — 60 seconds. Then we reveal and sharpen it together.

RECALL · REVEAL

8:23

The value shifts: from cost saved to insight gained

Rule-based bot → cost

Does what people already do, faster and cheaper. The business case is labour-hours removed.

Intelligent automation → insight

Does what people cannot: reads every contract and message, at a scale and speed no team could reach. The prize is the pattern that only shows up when you read all of it, not a sample.

The board case is not a cheaper back office. It is a company that sees and decides at a scale it could not reach before. Iansiti and Lakhani call it the AI factory.

RECONNECT · YOUR TURN

8:24

Last week you built a bot and broke it. What broke — and what did breaking it prove?

Cast your mind back to Lab 2. The bot ran fine, then hit the rows you seeded, and failed.

What broke

Where in the process did the bot fail — and what kind of step was it?

What it proved

What did that failure say about the limit of rules — and what has to take that step instead?

Talk it out in your team — 60 seconds. Then we reveal and sharpen it together.

You integrated it, automated it, broke it — now make it intelligent

Module 1 — integrate

The integration backbone: one connected set of systems, one source of truth. Everything runs on it.

Module 2 — automate

RPA on the repeatable, rules-based work. In Lab 2 you built a working bot — and broke it at the judgment step.

Module 3 — make intelligent

A machine takes the step the rule couldn't — and soon pursues a goal on its own.

You build above what you have, not instead of it. Module 3 moves the Customer thread to the front, though today's case lands first in the back office.

8:30

One programme, three threads — Customer leads now

Customer

Serving the person as one promise. Module 3 leads here — agents land first at service, fulfilment, dispute resolution.

Operations & Supply Chain

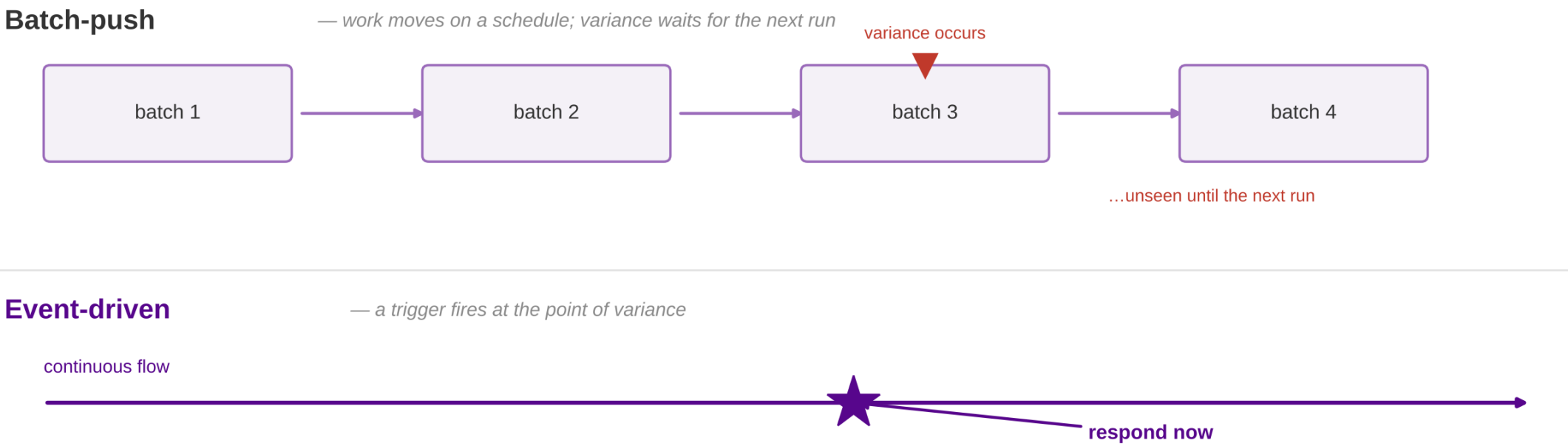
Planning, inventory, synchronisation. It led Module 2 — and today's bank case proves the mechanism here first.

Cash Management

How fast and reliably the backbone turns activity into cash. A faster, surer decision speeds the cash it governs.

Tag your judgment step to the thread it moves. A board funds a thread outcome, not a capability or a bot.

Stop pushing work in batches; react to events



Add a smart model to a batch-world job and you get a faster version of the same dysfunction. The gain lands only when the system acts at the moment the exception occurs.

STORY · THE CASE

8:36

A judgment step at industrial scale

JPMorgan COiN — Contract Intelligence: a machine-learning system that reads commercial loan agreements.

The work

Every commercial loan agreement must be read and interpreted — its terms, obligations, exceptions. Reading a clause is judgment, and no two contracts are quite alike.

The cost of doing it by hand

≈ 360,000 lawyer- and loan-officer-hours a year across ~12,000 contracts — and misreadings became loan-servicing errors downstream.

Son, Bloomberg (2017); open access via ABA Journal (2017).

Unstructured input, a rulebook that can't cover every contract, a costly error when it's wrong. That is a judgment decision point, not clerk work.

STORY · YOU DECIDE

8:40

You run the bank's contract review. How do you automate it?

A — Rules-based checker

Match keywords against a fixed template. Fast to build, easy to explain.

B — A system that learns to read

Train on thousands of contracts; generalise to agreements it has never seen.

C — Keep it human

It's judgment — don't automate it. Pay the 360,000 hours.

A show of hands, no huddle — A, B, or C? One of these breaks the moment a contract is phrased in a way the template didn't foresee. Then we see what the bank did.

STORY · WHAT HAPPENED

8:43

It didn't check the contracts — it learned to read them

How COiN works

Built on machine learning, not a keyword template. It *“learns by ingesting data to identify patterns and relationships,”* generalising to agreements it hadn't seen.

What changed

Minutes per contract → **seconds.**

And because it read consistently, the servicing errors from human misreading went **down.**

Son, Bloomberg (2017).

That is the difference between automating a clerk and automating a reader. The two differ in kind, not degree. (The answer is B.)

STORY · THE TURN THAT MATTERS

8:47

When a machine takes the judgment step, the question changes

The question moves from “does it work?” to three governance questions:

Who is accountable?

If the model misreads a clause — the model, the team that trained it, the officer who relied on it?

What is transparent?

Can a non-technical person see how it reached its reading — and check it? (Explainability.)

Where does a human stay?

On the high-value, the unusual, the case the model flags uncertain. (Human-in-the-loop.)

Where the machine acts and where the human stays is the heart of the design. It is what your Board Memo has to get right.

A model is a capability; a platform makes it usable

ENTERPRISE-GRADE GOVERNANCE · security · access control · ethics & bias

Inputs

- Human-driven (plain-language ask)
- Auto-discovery (process mining)
- Business-reactive (exception patterns)

→

EXECUTION ENGINE

orchestration

turns a reading into action across your systems

→

Must-have characteristics

- Explainability
- Utility-like (plug & run)
- Observability
- Adaptability (feedback loop)

The model reads; the platform turns the reading into action, and keeps a human able to see and stop it. (Tella, Brinker and Pezzini, 2023.)

YOUR TEAM'S WORK · 8:55–9:25

Name the step that needs AI judgment

Two short team sprints produce a real deliverable: the one step in your process a rule can't hold, the value it carries, and where a human stays. → Board Memo §5, slide S8 of your board deck.

YOUR TEAM'S WORK · THE ASK

8:55

Three questions frame the work — these are your study questions

1 What can rules-based RPA not handle — and why does that boundary matter strategically?

2 How does intelligent automation shift the value from cost-saving to insight generation?

3 Using COiN: a rules-based checker vs. a genuine ML system — and what changes in governance?

Work them in your team; I come round to each. By 9:25 you leave with the setup for slide S8: the step, its value, and its human boundary.

SPRINT 1 · BY TEAM · 8:55–9:10

Find the one step where the rule runs out

On your Lab 1 / Lab 2 process map, hunt the judgment decision point — the step a rule can't hold.

START HERE Mark each decision point stable-rules or judgment. Pick the single clearest judgment step.

GO FURTHER Say why the rule fails there — unstructured input, or a case it can't foresee? Which ML family would read it?

OUTPUT: one judgment step, named, + one line on why rules fail there.

Close-out: each team names its step in a sentence. This is your Board Memo §5 starting line, and the target your Lab 3 agent will take.

SPRINT 2 · BY TEAM · 9:10–9:25

Make the case: the value, and where a human stays

For the step you just named: price its value by thread, and draw its human boundary.

The value — by thread

START HERE One line of value; name the thread it moves — Customer, Operations, or Cash.

GO FURTHER The insight a model-at-scale unlocks — why it's an investment, not an expense.

The human — the boundary

START HERE One case you'd route to a person — ambiguous, high-value, irreversible.

GO FURTHER Who's accountable if it's wrong, and what must the platform make visible?

OUTPUT: a value line (by thread) + a one-line human boundary. → Board Memo §5.

Close-out: two teams share. A board funds a thread outcome, and trusts a design that says where the human stays.

CLOSE-OUT · REPORT-OUT

9:25

Each team — two minutes on your reasoning

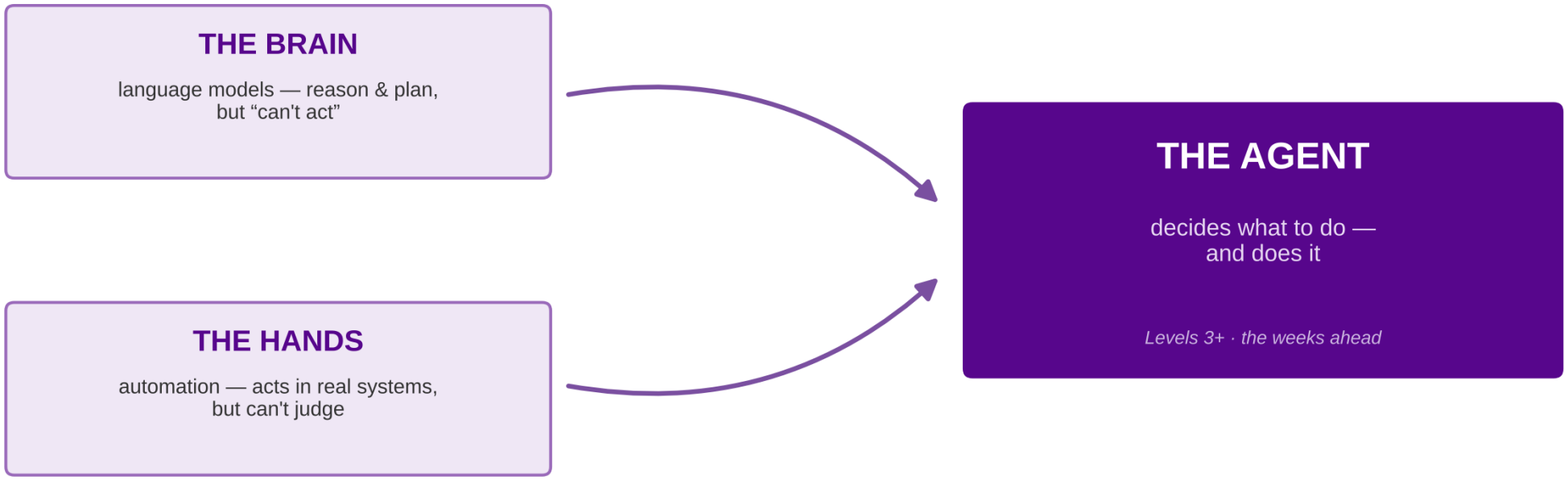
Walk us through your chain:

the step → **why rules fail** → **the value, by thread** → **where the human stays**

What leaves the room: your agent-design setup — Board Memo §5, slide S8 of the final board deck.

Two minutes per team, no slides. Just talk us through the chain. Then we look at where this is all heading, and our partners step in.

Agents are here because thinking and acting finally met



No single new invention made this happen. The thinking stream and the acting stream matured at the same moment, and could finally be joined (Bornet et al., 2025).

9:38

SPAR — the four capabilities you meet next week

Sense

take in the situation

Plan

work out the steps

Act

do them in real systems

Reflect

check its own work

Named today; taught next week; built in Lab 3. Read for the four parts, and for where Reflect has to sit if an agent is to catch its own mistakes.

INDUSTRY PANEL · YOUR TURN

9:43

Our guests are practitioners. What would you ask them?

Halfway through the course: integrate, automate, make intelligent, make agentic. Jot one question from anything that's puzzled you so far.

From the work

Something the ERP, automation, or AI material raised that you'd want a practitioner's read on.

From the career

How this actually plays out in their organisation — and what they look for in people who build it.

Bring one question. The sharpest come from what you've actually wrestled with, and from the capstone you're building.

INDUSTRY PANEL · MID-SEMESTER · 9:45–10:20

Halfway in — the people who do this for real

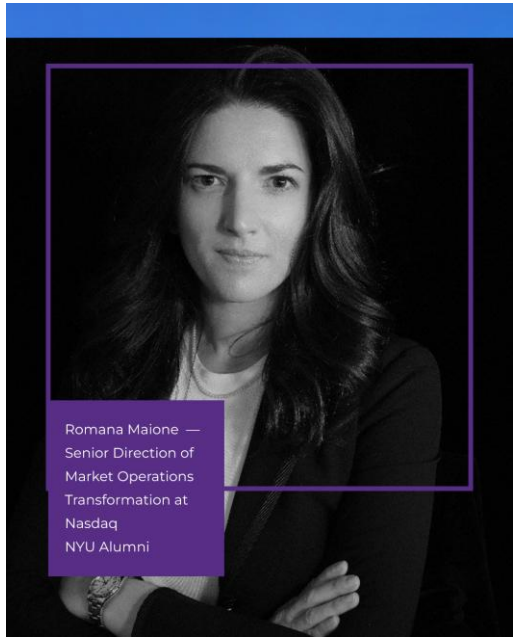
A step back across the whole arc: integrate, automate, make intelligent, make agentic. The partners who guest-lecture through the term, and who sit on the board panel your team presents to in December.

INDUSTRY PANEL · WHO'S HERE

9:46

Meet Our Industry Panel









These partners guest-lecture through the term and sit on your December board panel. Today they step back and take the whole journey with you, from the backbone to the agentic enterprise.

9:52

The questions on the table

1. Integration first: years into digital investment, where do backbone and data problems still bite?
2. Automate, intelligent, agentic — where is the real value today, and where is the hype?
3. Customer, operations, or cash: which thread is AI actually moving the needle on for you?
4. When a machine takes a decision a person used to make — how do you govern it, and what do you want from the engineers who build it?

Listen for where their line sits across the whole journey. It is the same judgment you'll defend to the board in December. Bring your own question too.

10:20

Before next week

Draft	The AI-judgment step + rationale → Board Memo §5 (Agent Design, slide S8). One step, named, with why rules fail there.
Read	The Week 8 study note
Keep evergreen	Your board deck current through S8 (setup). Add it while it's fresh.

You leave with the step an agent will take. Named and defended today; architected next week; built in Lab 3.

WEEK 7 · RPA → INTELLIGENT AUTOMATION

You named the step a machine should take.

Next · Week 8: SPAR — Inside the Mind of an AI Agent